

Natural Language Processing-16742

ITAI-2373

RT 2026 Section 01 3 Credits 01/12/2026 to 05/10/2026 Modified 01/07/2026

Our Vision

Houston City College will deliver relevant, high-quality education and training, ensuring success for all students, our community, economy and beyond. [About HCC \(https://www.hccs.edu/about-hcc/\)](https://www.hccs.edu/about-hcc/)

Course Meetings

Course Modality

Add Content Here

Meeting Days

Add Content Here

Meeting Times

Add Content Here

Meeting Location

Add Content Here

Lecture & Lab Exercise

Monday, 5:00 PM to 7:00 PM, Online

Monday 5-7pm CST

It is an OPTIONAL meeting. Teams link: [Online teams meeting_\(https://teams.microsoft.com/l/meetup-join/19%3Ameeting_ODZhMjY1OTctNjJiOS00NWY3LWJlODctZjEwMDVmZDc1MDAz%40thread.v2/0?context=%7B%22Tid%22%3A%22bedc536d-9380-4d10-9954-ddfc8f5b5bdd%22%2C%22Oid%22%3A%221b13416d-a3c3-4668-b535-a9aa656df92b%22%7D\)](https://teams.microsoft.com/l/meetup-join/19%3Ameeting_ODZhMjY1OTctNjJiOS00NWY3LWJlODctZjEwMDVmZDc1MDAz%40thread.v2/0?context=%7B%22Tid%22%3A%22bedc536d-9380-4d10-9954-ddfc8f5b5bdd%22%2C%22Oid%22%3A%221b13416d-a3c3-4668-b535-a9aa656df92b%22%7D)

Teams call will be recorded for viewing at your convenient time.

Welcome and Instructor Information

What's Exciting About This Course

"Natural Language Processing in 2026" represents a radical departure from traditional curricula, transforming students from passive observers of language theory into architects of intelligent, agentic systems. What makes this course truly electrifying is its refusal to treat AI as a "black box"; instead, it empowers robotics engineers to open the hood, teaching them how to optimize massive Large Language Models for the resource-constrained edge environments of the physical world. By moving beyond simple text generation to building self-correcting RAG pipelines and autonomous agents that can plan and execute tasks, the course bridges the critical gap between "chatting with a bot" and "commanding a robot." It is a comprehensive journey into the engineering reality of 2026, equipping students with the "Iron Triangle" mindset—balancing cost, latency, and accuracy—to deploy the next generation of safe, efficient, and reasoning-capable AI.

My Personal Welcome

Welcome to the course!

I'm Sri a Principal at Microsoft and Adjunct at HCC.

Preferred Method of Contact

Canvas Inbox

Online class discussions (Canvas)

Course Overview

This is an intermediate course on Natural Language Processing (NLP), which is a subfield of linguistics. Computer science, information engineering, and artificial intelligence (A. I.) concerned with the interactions between computers and human (natural) language. This course will give students an overview of the most critical pillars in A. I. and one of the most exciting and challenging areas of A. I. due to the challenges of language incorporate near linguistics, continuous processing, sound processing, and massive ambiguities of context/semantics/meaning. Additionally, studying A. I. such as Siri, Alexa, and Mycroft in-depth and the best-known NLP system that ranges from Hanson Robotics to Online Customer Service Systems.

Requisites

Lecture: 2

Lab: 4

Prerequisite(s): [MATH 1314 \(https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt7997\)](https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt7997), [ITAI 1370 \(https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt795\)](https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt795), [ITAI 1371 \(https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt5120\)](https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt5120), [ITAI 1372 \(https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt6877\)](https://catalog.hccs.edu/preview_program.php?catoid=14&poid=6582&returnto=1092#tt6877).

Core Curriculum Objectives (CCOs)

ITAI 2373 satisfies Component Area Option in the HCCS core curriculum. The HCCS Core Curriculum Committee has specified that the course address the following core objectives:

- **Critical Thinking:** Students will demonstrate the ability to engage in inquiry and analysis, evaluation and synthesis of information, and creative thinking by completing a written assignment such as a book report, research paper, or essay.
- **Communication Skills:** Students will demonstrate effective development, interpretation and expression of ideas through written, oral, and visual communication by completing a written assignment such as a book report, research paper, or essay.
- **Quantitative and Empirical Literacy:** Students will demonstrate the ability to draw conclusions based on the systematic analysis of topics using observation, experiment, and/or numerical skills by completing textbook reading assignments, completing assignments, and answering questions on quizzes and exams that pertain to Course Student Learning Outcome #2 below.

Student Learning Outcomes and Objectives

Learning Objectives for each CSLO are mapped within your Canvas Course

Departmental Practices and Procedures

As your Instructor, it is my responsibility to:

- Provide the grading scale and detailed grading formula explaining how student grades are to be derived
- Facilitate an effective learning environment through learner-centered instructional techniques
- Provide a description of assignments
- Inform students of policies such as attendance, withdrawal, tardiness, and making up assignments
- Provide the course outline and class calendar
- Arrange to meet with individual students as needed

As a student, it is your responsibility to:

- Attend class in person and/or online
- Participate actively by reading and reviewing course material, interacting with classmates, and responding promptly in your communication with your instructor
- Complete the required assignments and exams
- Ask for help when there is a question or problem
- Keep copies of all paperwork, including this syllabus, handouts, and all assignments
- Be aware of and comply with academic honesty policies in the [HCCS Student Handbook](#)

Program-Specific Student Success Information

Expect to spend at least twice as many hours per week outside of class as you do in class studying the course content. Additional time is required for the assignments. The assignments provided will help you practice the concepts discussed in class lectures and hone your hands-on skills. Successful completion of this course requires a combination of the following:

- Reading the materials provided from the Internet
- Critically viewing the videos provided from the Internet
- Attending class in person and/or online
- Working with your teammates to complete assignments
- Participating in class activities

There is no short cut for success in this course; it requires reading and studying the material and more importantly completing the assignments.

Instructional Materials and Resources

Instructional Materials

The [HCC Online Bookstore \(https://hccs.bncollege.com/shop/hccs-central/page/find-textbooks\)](https://hccs.bncollege.com/shop/hccs-central/page/find-textbooks) provides searchable information on textbooks for all courses. Check with your instructor before purchasing textbooks because the book might be included in your course fees.

Add Instructional Materials Here

Temporary Free Access to E-Book

Here is the link to get temporary free access to a digital version of the text for fourteen days:

<< [add link] >>

Other Instructional Resources

Courseware

Content here...

Course Requirements

Assignments, Exams, and Activities

Type	Weight	Topic	Notes
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Type	Weight	Topic	Notes
Written Assignment	15	Changes accordingly to the lecture topic	The assignment format will vary from written Word documents or PDF, PowerPoint presentations, or Multimedia submissions or text file or code.
Exams/Quizzes	40	key concepts covered in weekly modules	Separate multiple-choice, true/false, or short-answer questions, match the best answer, code related NLP focused patterns practices. LLM (with NLP based) advancements focused on the main ideas and theories discussed in the lectures.
In-Class Activities	5	Changes accordingly on the Lecture topic	This will be based on Lab exercises and ability to use GenAI tools with Colab and other IDEs.
Final Exam & Project	35	Final Project	In groups, or individual, students will create a proposal for integrating Gen AI and NLP learnings into a new or existing process within a chosen sector/Industry. This project will showcase the ability to innovate and apply theoretical knowledge to practical scenarios and provide value to the business/community/nation.
Extra Credit	5	Upload code to GitHub	For bonus point ! Keep building your AI portfolio by uploading the assignments from this course to GitHub

Grading Formula

Grade	Range	Notes
A	=>90	
B	80-89	
C	70-79	
D	60-69	
F	<=59	

* Instructor's Practices and Procedures

Incomplete Policy

Add Content Here

Missed Assignments/Make-Up Policy

Make-up exams will be given only in cases of extenuating circumstances. Extenuating circumstances are unexpected and unavoidable situations such as hospitalization or auto accident. They don't include forgetting about the date of the exam, busy work schedule, etc. You would need to provide documentation to your instructor as soon as possible after the missed assignment/assessment for consideration. Extenuating circumstances will be evaluated by your instructor on a case by case basis. It is your responsibility to contact your instructor with documentation of your situation as soon as possible, schedule a makeup exam, and submit the proper documentation to the departmentAdd Content Here

Academic Integrity

Scholastic Dishonesty will result in a referral to the Dean of Student Services.

See the link below for details. Students may ask questions to other students, to me, or to anyone else. This is how we learn, and I encourage it. HOWEVER, all work must be started and completed in its entirety with your group or on your own. If it is found that students Groups are sharing the same files, and then making minor changes to submit the work as their own, students will receive a 0 on the assignment and be referred to the Dean of Student Services

Here's the link to the HCC information about academic integrity (Scholastic Dishonesty and Violation of Academic Scholastic Dishonesty and Grievance):

<https://www.hccs.edu/studentprocedures> (<https://www.hccs.edu/studentprocedures>).

Here's the link to the HCC information about academic integrity (Scholastic Dishonesty and Violation of Academic Scholastic Dishonesty and Grievance): [HCC Student Procedures](https://www.hccs.edu/studentprocedures) (<https://www.hccs.edu/studentprocedures>).

Attendance Procedures

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Student Conduct

Add Content Here

Instructor's Course-Specific Information

TO ACCESS OUR COURSE go to <http://eagleonline.hccs.edu> Alternatively, you may go to <http://www.hccs.edu>. If you don't see the MyEagle Student Sign-in link at the top of your screen, scroll down to College Links (bottom right column) to select Student Sign-in, and then select Eagle Online Canvas.

READ THE MAIN PAGE OF OUR CLASS WEBSITE and then click on MODULES. Start with the first module, read each link in the module, and proceed to the second

Always check the class website and your student email AT LEAST ONCE a DAY

DOWNLOAD APPROPRIATE SOFTWARES FOR THE COURSE INCLUDING MS OFFICE IF you do

not already have Word, Excel and POWERPOINT PRINT OUT THE COURSE CALENDAR found in this syllabus.

THE EAGLE ONLINE USER ID will be your HCC User ID (sometimes referred to as the "W" number). All HCC students have a unique User ID. If you do not know your User ID, you can look it up by visiting the HCC home page. YOU WILL NEED TO KNOW THIS USER ID AND PASSWORD

EAGLE ONLINE TECHNICAL HELP Go to <http://hccs.edu/online> and click on Technical Support and then on Eagle Online Technical Support. It lists the support methods available:

The recommended way to get support is to complete the form under the Student Help Form

Technical Support number: 713-718-5275, option 3 IT Password Reset number: 713-718-8800, option 1

Devices

Add Content Here

Faculty Statement about Student Success

Expect to spend at least twice as many hours per week outside of class as you do in class studying the course content. Additional time will be required for the assignments. The assignments provided will help you practice the concepts discussed in class lectures and hone your hands-on skills.

Successful completion of this course requires a combination of the following:

Reading the material Participating in class activities and Completing assignments

Faculty-Specific Information Regarding Canvas

This course section will use [HCC Eagle Online - Canvas \(https://eagleonline.hccs.edu\)](https://eagleonline.hccs.edu) to supplement in-class assignments, exams, and activities.

HCCS Open Lab locations may be used to access the Internet and Canvas. For best performance, Canvas should be used on the current or first previous major release of Chrome, Firefox, Edge, or Safari. Because it's built using web standards, Canvas runs on Windows, Mac, Linux, iOS, Android, or any other device with a modern web browser.

Canvas only requires an operating system that can run the latest compatible web browsers. Your computer operating system should be kept up to date with the latest recommended security updates and upgrades.

Statement Regarding Classroom Environment

Add Content Here



HCC Policies and Information

HCC Grading System

HCC uses the following standard grading system:

Grade	Grade Interpretation	Grade Points
A	Excellent (90-100)	4
B	Good (80-89)	3
C	Fair (70-79)	2
D	Passing (60-69), except in developmental courses.	1
F	Failing (59 and below)	0
FX	Failing due to non-attendance	0
W	Withdrawn	0
I	Incomplete	0
AUD	Audit	0
IP	In Progress. Given only in certain developmental courses. A student must re-enroll to receive credit.	0
COM	Completed. Given in non-credit and continuing education courses.	0

Link to Policies in Catalog and Student Handbook

Here's the link to the HCC Catalog and Student Handbook: [HCC Catalog \(https://catalog.hccs.edu/\)](https://catalog.hccs.edu/)

In it you will find information about the following:

- Academic Information
- Academic Support
- Attendance, Repeating Courses, and Withdrawal
- Career Planning and Job Search
- Childcare
- Ability Support Services
- Electronic Devices
- Equal Educational Opportunity
- Financial Aid TV (FATV)

- General Student Complaints
- Grade of FX
- Incomplete Grades
- International Student Services
- Health Awareness
- Libraries/Bookstore
- Police Services & Campus Safety
- Student Life at HCC
- Student Rights and Responsibilities
- Student Services
- Testing
- Transfer Planning
- Veteran Services

Link to HCC Academic Integrity Statement

[HCC Student Conduct \(https://www.hccs.edu/student-conduct\)](https://www.hccs.edu/student-conduct) (scroll down to subsections)

Campus Carry Link

Here's the link to the HCC information about Campus Carry:

[HCC Campus Carry \(https://www.hccs.edu/campuscarry\)](https://www.hccs.edu/campuscarry)

HCC Email Policy

When communicating via email, HCC requires students to communicate only through the HCC email system to protect your privacy. If you have not activated your HCC student email account, you can go [to HCC Eagle ID \(https://www.hccs.edu/email\)](https://www.hccs.edu/email) and activate it now. You may also use Canvas Inbox to communicate.

Office of Equal Opportunity and Title IX

Use the following link to access the HCC Office of Equal Opportunity and Title IX:

[HCC OEO Title IX \(https://www.hccs.edu/oeotix\)](https://www.hccs.edu/oeotix)

Ability Services

HCC strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on your disability (including long and short term conditions, mental health, chronic or temporary medical conditions), please meet with a campus Abilities Counselor as soon as possible in order to establish reasonable accommodations. Reasonable accommodations are established through an interactive process between you, your instructor(s) and Ability Services. It is the policy and practice of HCC to create inclusive and accessible learning environments consistent with federal and state law. For more information, please go to the following website: [HCC Accommodations \(https://www.hccs.edu/accommodations\)](https://www.hccs.edu/accommodations)

Title IX

Houston City College is committed to cultivating an environment free from inappropriate conduct of a sexual or gender-based nature including sex discrimination, sexual assault, sexual harassment, and sexual violence. Sex discrimination includes all forms of sexual and gender-based misconduct and violates an individual's fundamental rights and personal dignity. Title IX prohibits discrimination on the basis of sex-including pregnancy and parental status in educational programs and activities. If you require an accommodation due to pregnancy please contact an Abilities Services Counselor. The Director of EEO/Compliance is designated as the Title IX Coordinator and Section 504 Coordinator. All inquiries concerning HCC policies, compliance with applicable laws, statutes, and regulations (such as Title VI, Title IX, and Section 504), and complaints may be directed to:

Sandra Jacobson, J.D., M.Ed., SHRM-SCP

Interim Director of EEO and Compliance/Title IX and ADA/Section 504 Coordinator

Office of Equal Opportunity and Title IX

3100 Main, 7th Floor

Houston, TX 77002

(713) 718-8271

hcc.oeotix@hccs.edu (<mailto:hcc.oeotix@hccs.edu>)

[HCC OEO Title IX Website](https://www.hccs.edu/oeotix) (<https://www.hccs.edu/oeotix>)

Mandatory Reporters

Under Texas Education Code 51.252 (formerly known as Senate Bill 212), HCC Instructors are mandatory reporters of sexual harassment, dating violence (domestic violence), sexual assault, and stalking. All instructors are required by law to report to the College's Title IX coordinator or Deputy Title IX coordinator all reports disclosed to them relating to sexual harassment, dating violence (domestic violence), sexual assault, and stalking alleged to have been committed by or against a person who was a student enrolled at or an employee of the institution at the time of the incident. Instructors are required by law to include all the information they know about the incident, including the name of the student(s), in the report to the College's Title IX coordinator or deputy Title IX coordinator.

Office of the Dean of Students

Contact the office of the Dean of Students to seek assistance in determining the correct complaint procedure to follow or to identify the appropriate academic dean or supervisor for informal resolution of complaints.

[HCC Dean of Students](https://www.hccs.edu/about-hcc/procedures/student-rights-policies--procedures/student-complaints/speak-with-the-dean-of-students/) (<https://www.hccs.edu/about-hcc/procedures/student-rights-policies--procedures/student-complaints/speak-with-the-dean-of-students/>)

Student Success

Expect to spend at least twice as many hours per week outside of class as you do in class studying the course content. Additional time will be required for written assignments. The assignments provided will help you use your study hours wisely. Successful completion of this course requires a combination of the following:

- Reading the textbook
- Attending class in person and/or online
- Completing assignments
- Participating in class activities

There is no short cut for success in this course; it requires reading (and probably re-reading) and studying the material using the course objectives as a guide.

Canvas Learning Management System

Canvas is HCC's Learning Management System (LMS), and can be accessed at the following URL:

[HCC Eagle Online - Canvas \(https://eagleonline.hccs.edu\)](https://eagleonline.hccs.edu)

HCCS Open Lab locations may be used to access the Internet and Canvas. For best performance, Canvas should be used on the current or first previous major release of Chrome, Firefox, Edge, or Safari. Because it's built using web standards, Canvas runs on Windows, Mac, Linux, iOS, Android, or any other device with a modern web browser.

Canvas only requires an operating system that can run the latest compatible web browsers. Your computer operating system should be kept up to date with the latest recommended security updates and upgrades.

HCC Online Information and Policies

Here is the link to information about HCC Online classes, which includes access to the required Online Information Class Preview for all fully online classes: [HCC Online \(https://www.hccs.edu/online\)](https://www.hccs.edu/online).

Scoring Rubrics, Sample Assignments, etc.

Look in Canvas for the scoring rubrics for assignments, samples of class assignments, and other information to assist you in the course: [HCC Eagle Online - Canvas \(https://eagleonline.hccs.edu\)](https://eagleonline.hccs.edu).

Instructor and Student Responsibilities

As your Instructor, it is my responsibility to:

- Provide the grading scale and detailed grading formula explaining how student grades are to be derived
- Facilitate an effective learning environment through learner-centered instructional techniques
- Provide a description of any special projects or assignments
- Inform students of policies such as attendance, withdrawal, tardiness, and making up assignments
- Provide the course outline and class calendar that will include a description of any special projects or assignments
- Arrange to meet with individual students during office hours, and before and after class as required

As a student, it is your responsibility to:

- Attend class in person and/or online
- Participate actively by reviewing course material, interacting with classmates, and responding promptly in your communication with me
- Read and comprehend the textbook
- Complete the required assignments and exams
- Ask for help when there is a question or problem
- Keep copies of all paperwork, including this syllabus, handouts, and all assignments
- Be aware of and comply with academic honesty policies in the [HCCS Student Handbook](https://www.hccs.edu/studenthandbook) (<https://www.hccs.edu/studenthandbook>).

Sensitive or Mature Course Content

In this college-level course, we may occasionally discuss sensitive or mature content. All members of the classroom environment, from your instructor to your fellow students, are expected to handle potentially controversial subjects with respect and consideration for one another's varied experiences and values.

EGLS3

The EGLS³ ([Evaluation for Greater Learning Student Survey System](https://hccs.campuslabs.com/courseeval/) (<https://hccs.campuslabs.com/courseeval/>)) will be available for most courses near the end of the term but before finals. Results are confidential and will be available to faculty and division chairs after the end of the term, after grades are due. Confidential means that responses to the survey will not be associated with your identity when provided to the faculty.

[Link to HCC EGLS3](https://www.hccs.edu/egls3) (<https://www.hccs.edu/egls3>).

Housing and Food Assistance for Students

If you are experiencing any hardship related to food, shelter, mental health, or other basic needs areas, please visit the Basic Needs page for resources ([HCC Cares](https://www.hccs.edu/cares) (<https://www.hccs.edu/cares>)). You have the option to take the Basic Needs Questionnaire and ask to be contacted by a counselor for additional assistance or support ([HCC Basic Needs](https://www.hccs.edu/basicneeds) (<https://www.hccs.edu/basicneeds>)). Furthermore, please notify the professor if you are comfortable doing so.

Student Resources

Tutoring

HCC provides free and convenient academic support, in a large variety of subjects, to HCC students in both an online environment and in-person on campus. Tutoring is provided by HCC personnel in order to ensure that it is appropriate. Visit the HCC Tutoring Services website for more information at [HCC Tutoring](https://www.hccs.edu/tutoring) (<https://www.hccs.edu/tutoring>).

Libraries

The HCC Library System consists of 9 libraries and 6 Electronic Resource Centers (ERCs) that are inviting places to study and collaborate on projects. Librarians are available both at the libraries and online to show you how to locate and use the resources you need. The libraries maintain a large selection of electronic resources as well as collections of books, magazines, newspapers, and audiovisual materials. The portal to all libraries' resources and services is the HCCS library web page at [HCC Library \(https://library.hccs.edu\)](https://library.hccs.edu).

Supplementary Instruction

Supplemental Instruction is an academic enrichment and support program that uses peer-assisted study sessions to improve student retention and success in historically difficult courses. Peer Support is provided by students who have already succeeded in completion of the specified course, and who earned a grade of A or B. Find details at [HCC Supplemental Instruction \(https://www.hccs.edu/supplemental-instruction\)](https://www.hccs.edu/supplemental-instruction).

Resources for Students:

[HCC COVID-19 Information for Students \(https://www.hccs.edu/covid19students\)](https://www.hccs.edu/covid19students)

Basic Needs Resources:

[HCC Cares Basic Needs Resources \(https://www.hccs.edu/support-services/counseling/hcc-cares/basic-needs-resources/\)](https://www.hccs.edu/support-services/counseling/hcc-cares/basic-needs-resources/)

Student Basic Needs Application:

[HCC Basic Needs \(https://www.hccs.edu/basicneeds\)](https://www.hccs.edu/basicneeds)

COVID-19

Here's the link to the HCC information about COVID-19:

[HCC COVID-19 \(https://www.hccs.edu/covid-19\)](https://www.hccs.edu/covid-19)

Instructional Modalities

HCC offers many options for learning modalities and information may be obtained at [HCC College Your Way \(https://www.hccs.edu/campaigns/college-your-way/\)](https://www.hccs.edu/campaigns/college-your-way/).

All students, regardless of their selected learning options, will still have access to the support available to help with their success, including tutoring, student life, basic needs support, career and employment services, counseling and ability services, and supplemental instruction. HCC also provides financial aid options.

In-Person (P)

In-Person gives students the opportunity to participate in class in person.

Online on a Schedule (WS)

Online on a Schedule allows students to take classes online, but at scheduled dates and times. Instead of visiting the campus for class, students log into the class online at the specified time they selected during registration. This gives students a safe and flexible learning option that allows for more interaction with professors and classmates without coming to campus. Access to a computer and WIFI are necessary to complete this class successfully.

Online Anytime (WW)

Online Anytime gives students the flexibility to complete coursework throughout the semester at times that works best for them. When a student enrolls in Online Anytime at the beginning of a semester, the course instructor will share pre-determined deadlines and requirements, along with the syllabus and full details of all assignments that must be completed to successfully finish the course. This option-most like a traditional online course -gives students the flexibility of completing coursework without visiting the campus. Access to a computer and WIFI is necessary to complete this class successfully.

Hybrid (H)

Hybrid provides students the opportunity to attend a lecture class meeting half-time in person and half-time remotely.

Hybrid Lab (HL)

Hybrid Lab-Based courses will be conducted in person and remotely, allowing HCC students access to essential hands-on, skills-based learning experiences needed for success.

Copyright Statement

In order to uphold the integrity of the academic environment and protect and foster a cohesive learning environment for all, HCC prohibits the unauthorized use of course materials. Materials shared in this course are based on my professional knowledge and experience as an instructor and are presented in an educational context for the students in the course. Authorized use of course materials is limited to personal study or educational uses. Material should not be shared, distributed, or sold outside the course without permission. Students are also explicitly forbidden in all circumstances from plagiarizing or appropriating course materials. This includes but is not limited to publicly posting quizzes, essays, or other materials. This prohibition extends not only during this course, but after. Sharing of the materials in any context will be a violation of the HCC Student Code of Conduct and may subject the student to discipline, as well as any applicable civil or criminal liability. Consequences for unauthorized sharing, plagiarizing, or other methods of academic dishonesty may range from a 0 on the specified assignment and/or up to expulsion from Houston City College. Questions about this policy may be directed to me, your instructor, or to the Manager of Student Conduct and Academic Integrity.

Unauthorized Disclosure

"Unauthorized disclosure" occurs when any student provides instructional materials and/or assessments to other students in violation of a clear prohibition by the instructor. Examples include: posting assessment items to online sites such as Chegg or CourseHero; asking exam questions in forums like Reddit or Yahoo Answers; discussions of confidential question using Wechat or GroupMe, etc.

Our Vision

Houston City College will deliver relevant, high-quality education and training, ensuring success for all students, our community, economy and beyond.



Course Calendar

Syllabus Modifications

The instructor reserves the right to modify the syllabus at any time during the semester and will promptly notify students in writing, typically by e-mail, of any such changes.

When	Topic	Notes
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When	Topic	Notes																																										
		<table><tr><th>Week</th><th>Dates</th><th>Module</th><th>Topic</th><th>Lab / Assignment</th><th>Due Date</th></tr><tr><td>1</td><td>Jan 12 – 18</td><td>Mod 0 & 1</td><td>Course Intro & The LLM-First World Pre-Transformer vs. Post-Transformer NLP</td><td></td><td>Mon, Jan 19* (MLK Day)</td></tr><tr><td>2</td><td>Jan 19 – 25</td><td>Mod 2</td><td>Text Representation From TF-IDF to Contextual Embeddings</td><td></td><td>Mon, Jan 26</td></tr><tr><td>3</td><td>Jan 26 – Feb 1</td><td>Mod 3</td><td>The Engine: Transformers Self-Attention, Encoders, & Decoders</td><td></td><td>Mon, Feb 2</td></tr><tr><td>4</td><td>Feb 2 – 8</td><td>Mod 4</td><td>Prompt Engineering Zero-shot, CoT, & DSPy</td><td></td><td>Mon, Feb 9</td></tr><tr><td>5</td><td>Feb 9 – 15</td><td>Mod 5</td><td>Fine-Tuning (Specialization) NER & Classification on the Edge</td><td></td><td>Mon, Feb 16</td></tr><tr><td>6</td><td>Feb 16 – 22</td><td>Mod 6</td><td>Generation & Summarization Decoding Strategies & LoRA/PEFT</td><td></td><td>Mon, Feb 23</td></tr></table>	Week	Dates	Module	Topic	Lab / Assignment	Due Date	1	Jan 12 – 18	Mod 0 & 1	Course Intro & The LLM-First World Pre-Transformer vs. Post-Transformer NLP		Mon, Jan 19* (MLK Day)	2	Jan 19 – 25	Mod 2	Text Representation From TF-IDF to Contextual Embeddings		Mon, Jan 26	3	Jan 26 – Feb 1	Mod 3	The Engine: Transformers Self-Attention, Encoders, & Decoders		Mon, Feb 2	4	Feb 2 – 8	Mod 4	Prompt Engineering Zero-shot, CoT, & DSPy		Mon, Feb 9	5	Feb 9 – 15	Mod 5	Fine-Tuning (Specialization) NER & Classification on the Edge		Mon, Feb 16	6	Feb 16 – 22	Mod 6	Generation & Summarization Decoding Strategies & LoRA/PEFT		Mon, Feb 23
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		7	Feb 23 – Mar 1	Mod 7	Retrieval Augmented Generation (RAG) Vector DBs & Grounding		Mon, Mar 2
		8	Mar 2 – 8	Mod 8	Chatbots & Agents Memory, Tools, and Agent Frameworks		Sun, Mar 8* (Due before Break)
		--	Mar 9 – 15	BREAK	SPRING BREAK	No Classes	No Due Dates
		9	Mar 16 – 22	Mod 9 & 10	Syntax & Topic Modeling Knowledge Graphs & Semantic Clustering		Mon, Mar 23
		10	Mar 23 – 29	Mod 11	Machine Translation NMT & Multilingual Models		Mon, Mar 30
		11	Mar 30 – Apr 5	Mod 12	The LLM Ecosystem & MLOps Quantization, Edge AI, & Cost		Tue, Apr 7* (Adjusted for Spring Holiday)
		12	Apr 6 – 12	Mod 14	Ethics, Safety & Alignment Bias, Jailbreaking, & Red Teaming		Mon, Apr 13
		13	Apr 13 – 19	Mod 15	Multimodality & Future Trends Vision-Language Models (VLM)	Assignment: Final Project Proposal	Mon, Apr 20

When	Topic	Notes					
		14	Apr 20 – 26	Project	Final Project Sprint 1 Implementation & Prototyping	Progress Report: MVP Demo / Code Check-in	Mon, Apr 27
		15	Apr 27 – May 3	Project	Final Project Sprint 2 Evaluation & Refinement	Draft Report: Project Progress/Eval Metrics & Documentation	Mon, May 4
		16	May 4 – 10	Mod 16	Course Wrap-Up Final Presentations	Final Deliverable: Live Demo & Final Paper	Fri, May 8

Additional Information

Artificial Intelligence

Artificial Intelligence is the current, future discipline fostering the next generation of Computer Science, Networking, Cloud and High Performance Computing environments.

Degrees and Certificates

- Artificial Intelligence (A.I.) - AAS (Associate of Applied Science)

Degree program for Artificial Intelligence will prepare students for the following entry- level job roles:

- A.I. Application Developer
- A.I. Software Architect
- Artificial Intelligence Developer (software)

Employment

Examples of specific jobs held by AI professionals include:

- Software analysts and developers and programmers
- Computer scientists, computer engineers, cyber security specialists
- Algorithm specialists, Data Scientists, Data Engineers, Data Researchers
- Research scientists, engineering consultants, Cloud Computing Specialists
- Mechanical engineers and maintenance technicians.
- Manufacturing and electrical engineers. Software analysts and developers.

The estimated annual job openings is approx. 3,000+ jobs a year.

- Median Wages - \$79 hourly, \$92,000 - \$150,000 annually.
- Job Roles and the shift from Computer Software Developers to AI Developers.

Student Clubs

- Computer Science Association (CSA) club dit.csa@hccs.edu
- Women in Technology (WIT) club <https://hccs.presence.io/organization/women-in-technology>

Process for Expressing Concerns about the Course

If you have concerns about any aspect of this course, please reach out to your instructor for assistance first. If your instructor is not able to assist you, then you may wish to contact the Department Chair.

Liene Joshua Lo

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Office: C.152